resize themselves as well. However, as a result, when you create a *toplevel* (an empty window) container, you can place only one control directly inside it. To get around this, you place a *container* control inside the *toplevel* window, and add other controls inside the container. You create complex layouts by nesting containers one inside the other, as we will see in this tutorial. This usually entails some planning of what you want to have in your window—though, as you will soon realise, the initial steps are almost the same for many windows.

Before we begin, take a look at Figure 1, which is the login screen we intend to create. Notice that we want to have a menu bar, a notebook container control that will hold multiple 'screens', and a status bar. Now, let's start creating the window.

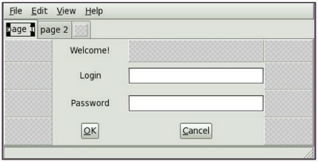

Figure 1: The target of our design

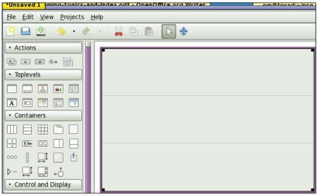

Figure 2: VBox placed in toplevel window

Toplevel window and VBox

In the palette at the left of the Glade window, you should see a section titled '*Toplevels*'. If you hover your mouse pointer above each of the buttons, you will see tool-tips, one of which says, 'Window'. Click that button, and a *toplevel* window is shown in the design area.

Since the *toplevel* window can contain only one control, we must add a container control. There are three containers that are frequently used as the first control in a *toplevel* window: the HBox, the VBox, and the Table. If you look at the '*Containers*' section of the Glade palette, you will find those (their tool-tips are 'Horizontal Box', 'Vertical Box' and 'Table', respectively). Each of these has a number of 'slots' or 'cells' into which you can place controls. When adding (packing) controls into a Horizontal Box/HBox, the widgets are inserted horizontally, from left to right, or right to left (depending on property settings, some of which may not be visible in Glade but can be set in code). In a VBox, the 'slots' are stacked vertically, so the controls placed in them appear one below the other. If you look at Figure 1 again, you will realise this is just what we need: a menu bar at the top, a notebook below it, and a status bar below that. We'll use the *Table* container later in this tutorial.

Let's place a VBox in the *toplevel* window: click the 'Vertical Box' button in the 'Containers' section, and then click on the *toplevel* window in the design area. In the pop-up dialogue box, set the number of items for the VBox to 3. The upper left part of your Glade window should now resemble Figure 2.

Naming controls

Some developers insist on giving a meaningful name to each and every control placed in a window. From the practical viewpoint, this can be tedious and time-consuming. We assign a different name to a control only if we intend to access that control in code. For example, if we want to set a property of a control, or retrieve some data from a control (like the text in a text entry), then giving the control a memorable name is a good idea. The code becomes more readable, and both the code and the UI design become easier to maintain.

Since we have only one window in this tutorial Glade project, we have left its name at the default. As far as the VBox is concerned, we do not intend to manipulate it in code—hence, the default name *vbox1* is good enough.

> **Note:** To change the *Name* property, and other properties of a control, you would use the *Properties* pane, at the bottom right of the Glade window; look under the *General* tab for the *Name* property. Look at the 'Control Properties' box now, to read about some commonly used properties.

Next, add a menu bar control, a notebook control (both are in the palette's '*Containers*' section), and a status bar control (in the 'Control and Display' section) to the top, middle and bottom slots in *vbox1*. You can later experiment with different settings in the *Packing*, *Common* and *General* tabs in the *Properties* pane, selecting different controls from the object tree above the *Properties* pane.

Laying out the login screen

Returning to the notebook widget, we will use the first page for our login screen. We need to place several widgets